

Multiple Choice (10 marks)

1. What is the force between two positive charges of 100μ coulombs and 200μ coulombs, respectively, when they are separated a distance of 3 centimeters?
 - (a) 5×10^5 nt
 - (b) 1×10^5 nt
 - (c) 7×10^5 nt
 - (d) 3×10^5 nt
 - (e) 6×10^5 nt
2. The direction of natural processes is from states of
 - (a) lower order to higher order
 - (b) chaos to equilibrium
 - (c) higher order to lower order
 - (d) disorder to organization
 - (e) equilibrium to higher order
3. Any atom that emits an alpha particle or beta particle
 - (a) changes its atomic number but remains the same element.
 - (b) becomes an atom of a different element always.
 - (c) remains the same element and isotope.
 - (d) becomes a different element with the same atomic mass.
 - (e) becomes an atom of a different element sometimes.
4. The separation of the headlights on a car is about 150 cm. When the car is 30 meters away, what is the separation of the images of the lights on the retina?
 - (a) 4 mm
 - (b) .833 mm
 - (c) 1 mm
 - (d) 1.5 mm
 - (e) 0.5 mm
5. A hoist raises a load of 330 pounds a distance of 15 feet in 5 seconds. At what rate in horsepower is the hoist working?
 - (a) 2.5 hp

- (b) 2.7 hp
- (c) 1.8 hp
- (d) 1.2 hp
- (e) 3.0 hp

True / False (10 marks)

Answer True or False

6. True or False: The work done to stretch the steel wire is 0.3 nt-m, as it requires significant energy for deformation.
7. True or False: The refractive index of the glass for a -10.00 diopter biconcave lens with 12 cm radii of curvature is 2.0.
8. True or False: To stop a disk rotating at 1200 rev/min in 3.0 minutes, a torque of -26 lb-ft is required.
9. True or False: The correct answer to 'Find the center of mass of a uniformly solid cone of base diameter $2a$ and height h ' is ' $\frac{3}{4}h$ '.
10. True or False: The speed of sound in air is significantly influenced by the air temperature, increasing as the temperature rises.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A curling stone of mass 1 slug slides across ice and comes to rest after 100 ft due to a kinetic friction force of $\frac{1}{2}$ lb. Calculate the initial velocity and time taken to stop, providing a detailed analysis of your approach.
12. Discuss how to calculate the magnitude of the Poynting vector for a light wave traveling in glass with an index of refraction of 1.5 and an electric field amplitude of 100 volts/meter.

End of Paper